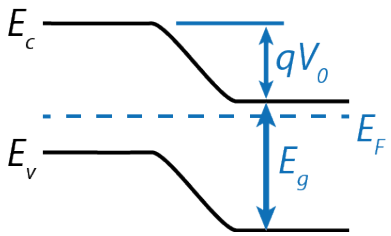
# Part I

## Question I-1

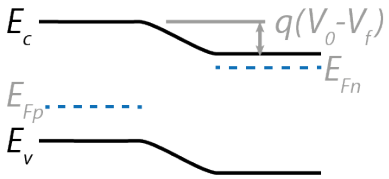
In the following we consider an abrupt Si p-n$^+$ junction with a more heavily doped n-side (i.e., $p_p \ll n_n$).

a) Sketch the energy band diagram for such a p-n$^+$ junction in thermal equilibrium, forward bias and reverse bias, respectively. Indicate the following: Band gap ($E_g$), Fermi level ($E_F$), and Contact potential ($V_0$).

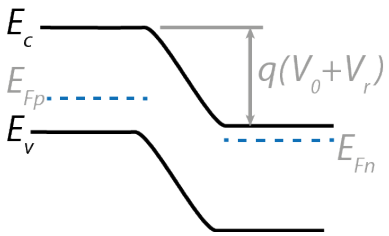



**Thermal equilibrium**

Note: $E_C$ and $E_V$ (as these are an inherent part of the energy band diagram) together with the parameters asked for in the a) should be indicated to get full score on this question. However, it is sufficient to indicate these in one of the three diagrams.




**Forward Bias**

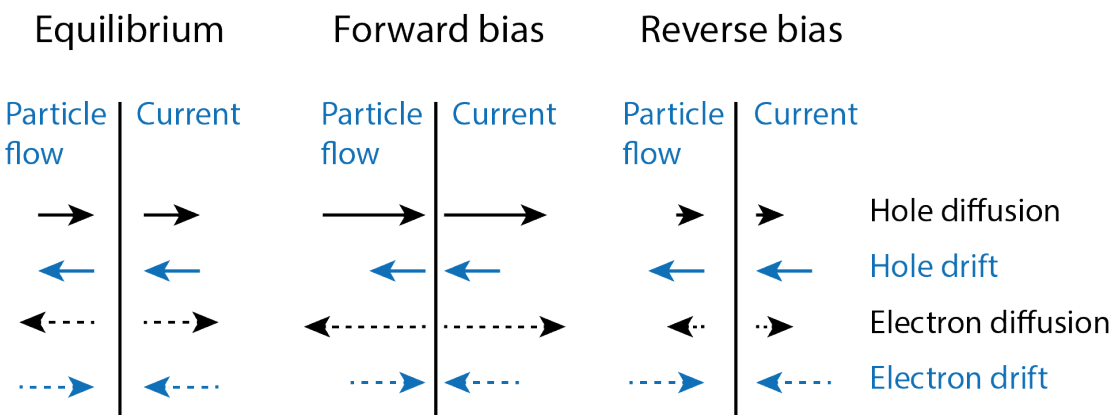



**Reverse Bias**

b) Describe the electron and hole drift and diffusion currents ( $I_n$ and $I_p$) under the three biasing conditions discussed in a). Indicate qualitatively the magnitude of the different current contributions and discuss these briefly. (*A sketch would be helpful, but not enough to get a full score.)*

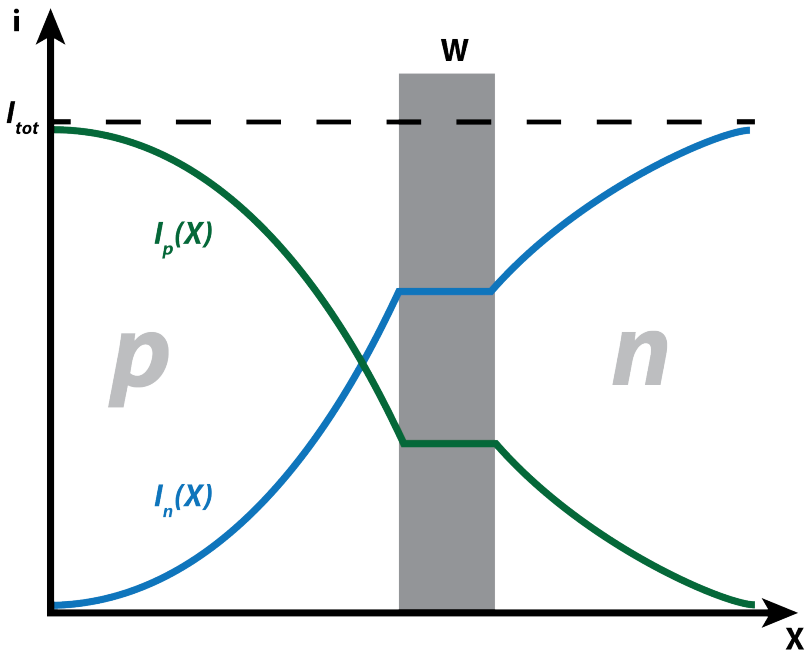
The *drift current* is almost independent of bias, as it is not limited by the electric field in the depletion region, but rather by availability of minority carriers at the edge of the depletion region. This availability is again limited by the thermal generation of EHP within one diffusion length of the edge of the depletion region.

The *diffusion current* on the other hand, for both holes and electrons, varies exponentially with the potential barrier at the junction and will therefor grow rapidly with a forward bias and be correspondingly reduced by a reverse bias.



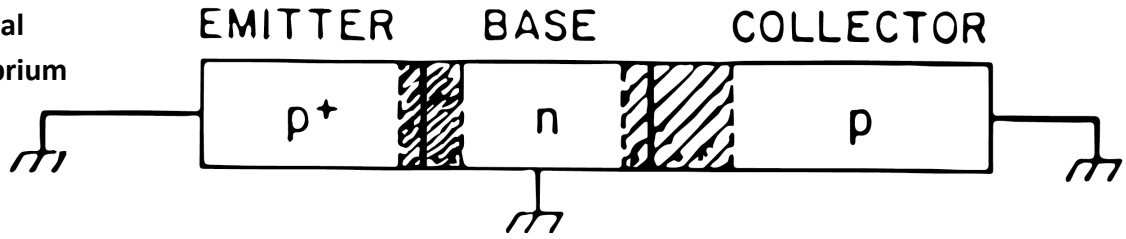

Note: The particle flow indicated in the above figure is not needed to get a full score on this question. As the p-n jct under consideration has a higher doping on the n-side. There will be an asymmetry in the hole and electron currents (see figure above). Failure to indicate this will result in a small deduction.

c) Show with an appropriate sketch how the electron and hole currents would vary through the p-n junction discussed in a) (that is, plot the current contributions vs. position). In addition to the electron and hole currents, indicate the total current as well as the depletion region in the plot. Assume that there is no recombination or generation in the depletion region and that both the p and the n-sides are long compared to the electron and hole diffusion lengths.

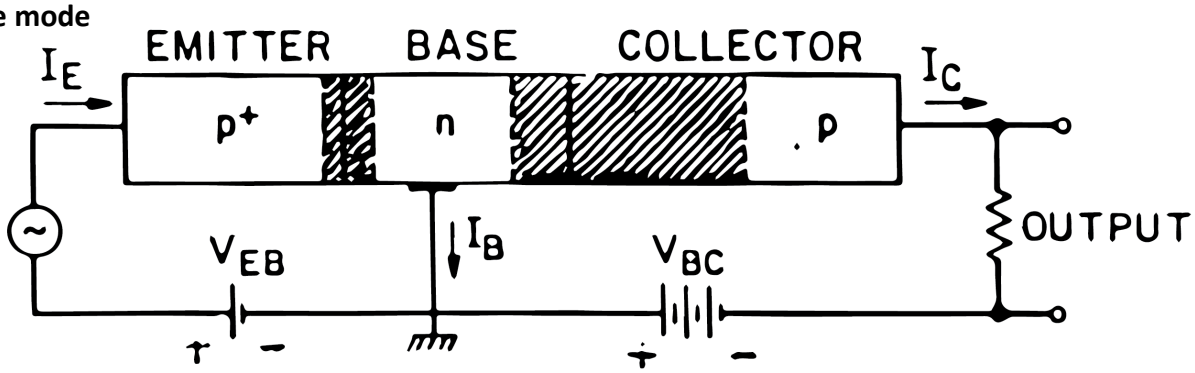


d) A *p-n-p* bipolar junction transistor (BJT) contains two p-n junctions. For a BJT with a doping profile of the *emitter, base and collector* such that $N_E > N_B > N_C$, sketch the device and indicate the depletion regions at the two junctions in thermal equilibrium (i.e., when the device is un-biased) and when it is biased in the normal active mode.

**Thermal equilibrium**

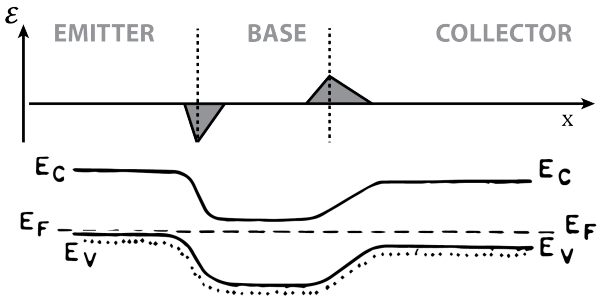



**Normal active mode**

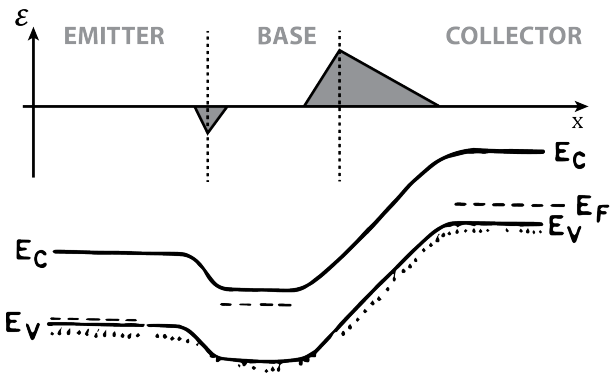



Note: The external circuit is not needed to get full score on this question.

e) Sketch the electric field distribution as well as the energy band diagram for the p-n-p BJT described in d) in thermal equilibrium and when biased in the normal active mode. Make use of the *depletion approximation* from ideal diode theory which assumes carrier depletion within the transition regions and neutrality outside these.




**Thermal equilibrium**




**Normal active mode**

Note: The position of the depletion regions should be compatible with the given doping profile and the maximum field and width of these regions should change in a sound manner when the BJT is biased as compared to the situation in thermal eq.

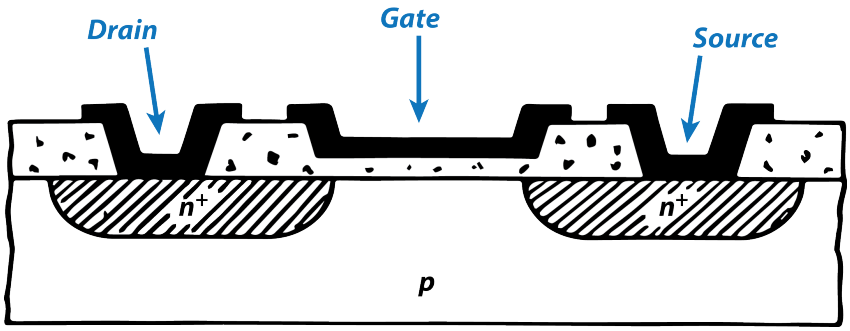
f) Assume that the BJT above can withstand very high temperatures without melting or otherwise be mechanically damaged. If the BJT was heated to >300°C, would it still work as a BJT? Explain why or why not.

Heating the device to temperatures above 300C would cause the intrinsic carrier levels in the emitter, base, and collector to become comparable to the extrinsic carriers (due to doping). Effectively the p-n junctions would not be p-n junctions anymore and we would in effect have three regions with close to intrinsic properties. Thus, the BJT would cease to operate as a BJT.

## Question I-2

In the following consider a Metal-oxide-semiconductor field-effect transistor (MOSFET) where the base is p-type and the source and drain contacts are doped $n^+$ doped.

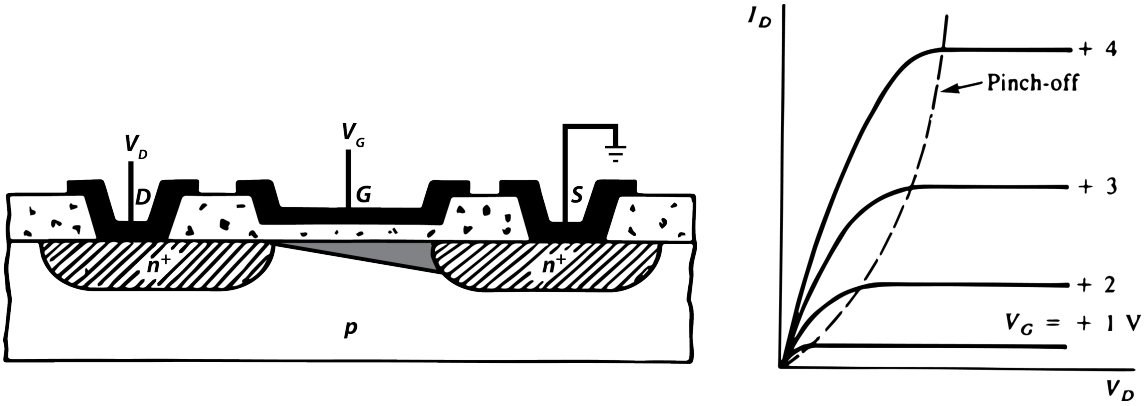
a) Sketch this MOSFET and indicate where we find the three terminal contacts. What do we call a MOSFET with this kind of doping and why are the source and drain contacts heavily doped?




A MOSFET with this kind of doping (i.e., a p-type substrate and n-type source and drain regions) is referred to as an n-channel MOSFET. This name is given due to the fact that the conductive channel which form under the gate and connect the source and drain regions will have n-type properties (i.e., the Fermi level will locally be above the intrinsic Fermi level). The contact regions are heavily doped to ensure an ohmic contact to the external circuit.

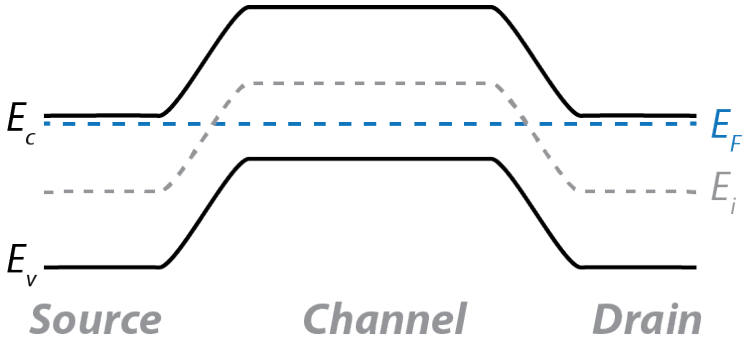
b) Explain the term *pinch-off* in relation to operation of the MOSFET described above. Also, sketch a typical $I_D$ - $V_D$ characteristics for such a device for a few different gate voltages and indicate the onset of pinch-off in this plot.

The drain-source voltage in a MOSFET will cause the potential to vary along the conducting channel. This will cause the effective bias between the substrate and the gate contact to be reduced near the drain contact. The reduced effective bias will cause the channel opening to be reduced near the drain and for a sufficiently large drain-source voltage the channel will be completely “pinched-off”. Thus, in pinch-off, one end of the conducting channel is restricted to zero by a depletion region, see sketch below.



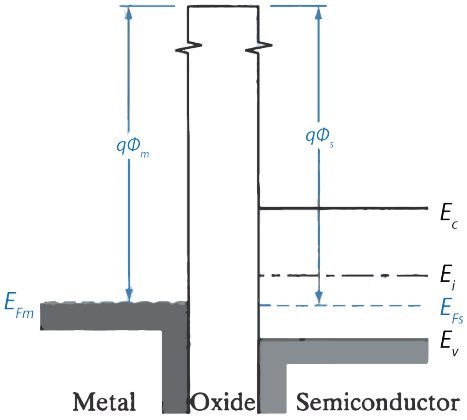

Note: A sketch is not a requirement for the explanation of "pich-off"

c) Draw the energy band diagram along the channel region for the MOSFET device described above in thermal equilibrium. That is, the band diagram going from source via the channel region to the drain. Explain how this diagram would change upon applying a gate bias.

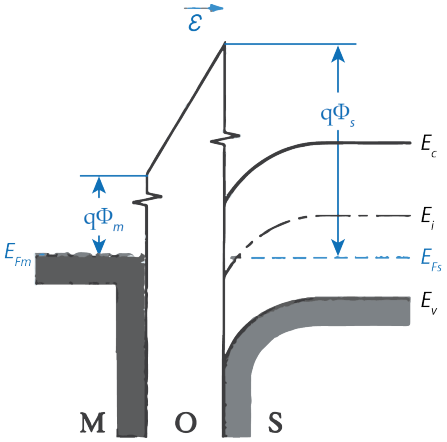



By applying a gate bias the potential barriers seen between source/drain and the channel region will be increased or reduced depending on the sign of the gate voltage. To open a channel in this device a positive gate bias must be applied. This would then reduce the two potential barriers and at a sufficiently strong bias a current can flow between source and drain.

d) The ideal MOS structure is essentially a plate capacitor in which one of the plates is a metal and the other a semiconductor. In this problem, assume the semiconductor to be p-type, as in the device described above. Sketch the energy band diagram for such a structure in thermal equilibrium. Assume that the modified work functions are different and let the work function difference be such that an inversion layer is formed for zero gate bias. Sketch the energy band diagram for this MOS structure in thermal equilibrium. Indicate the position of the Fermi level ($E_{FS}$) and the intrinsic Fermi level ($E_i$) in the semiconductor as well as the Fermi level ($E_{Fm}$) in the metal.




**Ideal MOS**




**MOS with different work functions**

Note: Answers with only one band diagram are also accepted as the question formulation was not quite clear.

# Part II

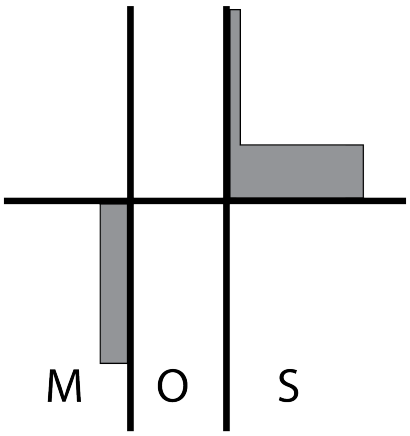
**Question II-1.**

Which is an important motive for replacement of $SiO_2$ by materials with a higher dielectric constant (also referred to as high-κ dielectrics) as the gate insulator in state-of-the-art Si MOSFETs?

a) To increase the carrier mobility at the semiconductor-oxide interface
b) To ensure enhancement mode operation, also in the case of lightly doped n-channel MOSFETs
**c) To reduce leakage current through the gate oxide**

**Question II-2.**

The sketch shows the space charge distribution for an ideal MOS capacitor under a certain bias. What kind of MOS structure and which bias condition is this?

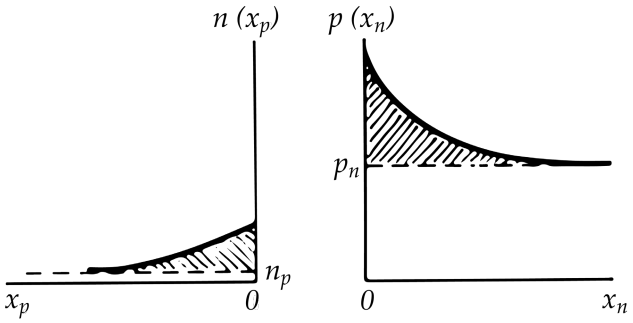



a) **p-type MOS under strong inversion**.
b) n-type MOS under strong inversion.
c) n-type MOS under accumulation.

**Question II-3.**

The figure below shows the minority carrier distribution in which of the following devices?

a) **A p-n junction under forward bias in which the p side is the most heavily doped**.
b) A p-n junction under forward bias in which the n side is the most heavily doped.
c) A p-n junction under reverse bias in which the p side is the most heavily doped.



**Question II-4.**

The carriers corresponding to the crosshatched region in the figure in Question II-3 constitutes the charge commonly referred to as:

a) The *depletion charge*.
**b) The *stored charge*.**
c) The *inversion charge*.

**Question II-5.**
Which properties are typically measured in the Hall-effect experiment?

a) The minority carrier mobility and concentration.
b) **The majority carrier mobility and concentration.**
c) The majority and minority carrier concentration.

**Question II-6**

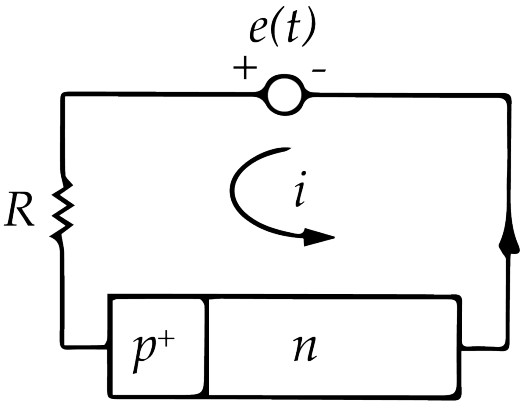
When measuring the I-V characteristics of a p-n junction in the lab, you observe that the reverse saturation current increases with increasing reverse bias (even before reaching breakdown). What could explain this behavior?

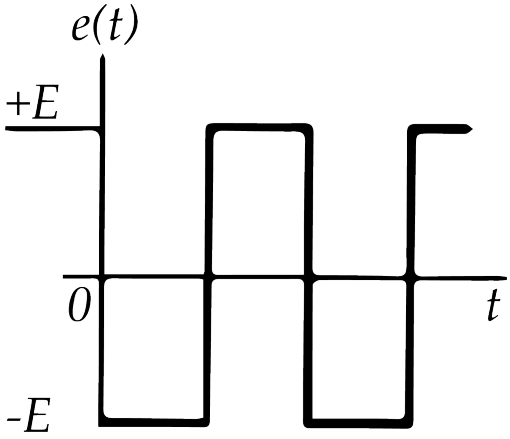
a) Ohmic effects in the p-n junction.
b) Diffusion of carriers across the depletion region.
**c) Generation of electron-hole-pairs within the depletion region.**

**Question II-7**

Considering the circuit sketched below containing a voltage source a resistor and a $p^+$-n junction. The voltage delivered from the voltage source as a function of time is showed in the plot to the right. If the current just before switching the voltage at $t = 0$ is equal to $I$, what is true for the current measured at a time $t = t_1$ when you assume that $t_1$ is small compared to both the period of the input signal from the voltage source and the minority carrier lifetime on the n-side of the $p^+$-n junction.

a) **The current at $t_1$ is approximately equal to $-I$.**
b) The current at $t_1$ is approximately equal to $I$.
c) The current at $t_1$ is approximately equal to $0$.



**Question II-8**

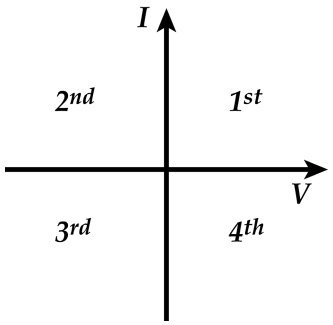
What is true for the ideal MOS structure at the onset of *strong inversion*?

a) **The conductivity is as high in the inversion channel as it is in the bulk substrate.**
b) The intrinsic Fermi level is equal to the Fermi level.
c) An inversion layer has just formed.

**Question II-9.**

In which quadrant of the I-V characteristic would a diode be operated when used as a photodetector?

a) 1st
**b) 3rd**
c) 4th




**Question II-10.**

In a p-n-p BJT, the hole transit time through the base region is denoted $\tau_t$ and the hole lifetime in the base is denoted $\tau_p$. How should the relation between the two be, in a device to be used for amplification?

a) $1 \gg \frac{\tau_p}{\tau_t}$
**b)** $\mathbf{1 \ll \frac{\tau_p}{\tau_t}}$
c) $1 \approx \frac{\tau_p}{\tau_t}$

# Part III

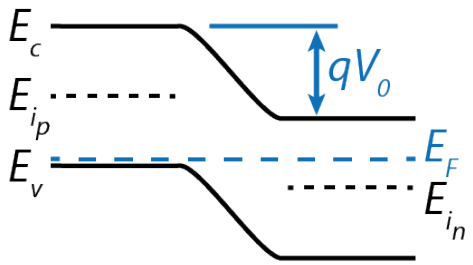
It might be necessary to make some sound assumptions when solving the questions in this section. To get a full score, you should clearly state important assumptions you make in order to arrive at your final answer.

Given: $\varepsilon_r(SiO_2) = 3.9$, $\varepsilon_r(Si) = 11.8$, $\varepsilon_0 = 8.85 \cdot 10^{-14}$ F/cm, $n_i(Si)^{300K} = 1.5 \cdot 10^{10}$ cm$^{-3}$

**Question III-1**

Consider a Si p$^+$-n junction, $10^{-2}$ cm$^2$ in area and with a base width (i.e., length of the n-side) of L=100µm. The dopant concentration on the n-side is $N_d = 10^{15}$ cm$^{-3}$ and the device is operated at 300K.

a) Sketch the band diagram for this diode in thermal equilibrium (qualitative accuracy is sufficient). Indicate the position of the intrinsic Fermi level in the neutral p- and n-regions as well as the contact potential.




b) Given a minority carrier lifetime of 1 µs and a diffusion coefficient of 10 cm$^2$/s for holes on the n-side, calculate the diode current at a forward bias of 0.6 V.

For a p$^+$-n junction we have that $N_a >> N_d$ and the forward current will be dominated by holes injected from the p-side into the n-side. As a forward bias of 0.6V gives $e^{qV/k_BT} >> 1$ at 300 K, the forward current may be expressed as:

$$I = qA\frac{D_p}{L_p}p_n e^{qV/k_BT}$$

At 300 K we may safely assume full ionization and find:

$$p_n = \frac{{n_i}^2}{n_n} = \frac{(1.5 \times 10^{10})^2}{1 \times 10^{15}} = 2.25 \times 10^5 cm^{-3}$$

$$L_p = \sqrt{D_p \tau_p} = \sqrt{10^{-5}} = 3.16 \times 10^{-3} cm^3$$

$$\boldsymbol{I = 1.6 \times 10^{-19}\,(10^{-2})\frac{10}{3.16 \times 10^{-3}}(2.25 \times 10^5)e^{\frac{0.6}{0.0259}} = 13.1\ mA}$$

c) What is the depletion region width, *W*, for this device at thermal equilibrium and at the onset of avalanche breakdown? The breakdown voltage may be found from Figure 1.

At the heavily doped p-side the Fermi-level will lie essentially at the valence band edge, that is, $E_g/2(\sim 0.55eV)$ bellow $E_i$. Assuming full ionization the Fermi-level position on the n-side is found from eq.3-25 as:

$$E_F - E_i = k_B T \, ln\frac{N_d}{n_i} = 0.0259 \ln\frac{10^{15}}{1.5 \times 10^{10}} = 0.288 \, eV$$

The contact potential can now be found as the difference between intrinsic Fermi-levels on the p- and n-side:

$$qV_0 = 0.55 + 0.288 = 0.838 \, eV$$

$$\boldsymbol{W_{eq}} = \left[\frac{2\epsilon V_0}{qN_d}\right]^{1/2} = \left[\frac{2 \times 11.8 \times 8.85 \times 10^{-14} \times 0.838}{1.6 \times 10^{-19} \times 10^{15}}\right]^{1/2} = 1.04 \times 10^{-4} cm \approx \boldsymbol{1.0} \, \mu\boldsymbol{m}$$

From Figure 3 we find the breakdown voltage $V_{br} \sim 350$ V for a $p^+$-n junction with $N_d = 10^{15}$ $cm^{-3}$. The width of the depletion region is found as:

$$\boldsymbol{W_{br}} = \left[\frac{2\epsilon V_{br}}{qN_d}\right]^{1/2} = \left[\frac{2 \times 11.8 \times 8.85 \times 10^{-14} \times 350}{1.6 \times 10^{-19} \times 10^{15}}\right]^{1/2} = 2.13 \times 10^{-3} cm \approx \boldsymbol{21} \, \mu\boldsymbol{m}$$

d) At which reverse bias would current breakthrough occur if the base width were to be restricted to 10µm. Find the numerical value and explain why the answer is different from that in c).

As this is a one-sided junction (i.e., $p^+$-n), we can assume that the depletion region lies entirely on the lightly doped n-side. If the base width is restricted to a width smaller than $W_{br}$ (see c) ), current breakthrough will occur when the depletion region fills the base. This situation is commonly referred to as "punch-through". Rewriting the equation used to find W in c) we find:

$$\boldsymbol{V_{p-t}} = \frac{qN_d {W_{br}}^2}{2\epsilon} = \frac{1.6 \times 10^{-19} \times 10^{15} \times (10^{-3})^2}{2 \times 11.8 \times 8.85 \times 10^{-14}} = \boldsymbol{76,6 \, V}$$

## Question III-2

Consider an n-channel MOSFET on a p-type Si substrate with a substrate doping of $N_a = 1 \cdot 10^{16}$ $cm^{-3}$ and with an $n^+$-polysilicon gate electrode. The gate oxide is a 10 nm thick layer of $SiO_2$ and the channel region is 1μm long and 25μm wide. There is an effective interface fixed charge of $Q_i = 2.5 \cdot 10^{-8}$ $C/cm^2$. The variation of the metal-semiconductor work function potential difference ($\Phi_{ms}$) with substrate doping concentration for $n^+$-poly-Si can be found from figure 2. Assume room temperature, T = 300 K and a channel mobility of $\bar{\mu}_n$= 200 $cm^2/V \cdot s$.

a) Calculate the width of the depletion region, W, at strong inversion.

$$\phi_F = \frac{kT}{q}\ln\left(\frac{N_a}{n_i}\right) = 0.0259\ln\left(\frac{1\times10^{16}}{1.5\times10^{10}}\right) = 0.347\ V$$

$$W_m = 2\left[\frac{\varepsilon_s\phi_F}{qN_a}\right]^{1/2} = 2\left[\frac{11.8\times8.85\times10^{-14}\times0.347}{1.6\times10^{-19}\times1\times10^{16}}\right]^{1/2}$$

$$\boldsymbol{W_m} = 3.0\times10^{-5}cm = \mathbf{0.30}\ \boldsymbol{\mu m}$$

b) Calculate the threshold voltage, $V_T$ and sketch the band diagram for the MOS capacitor at strong inversion.

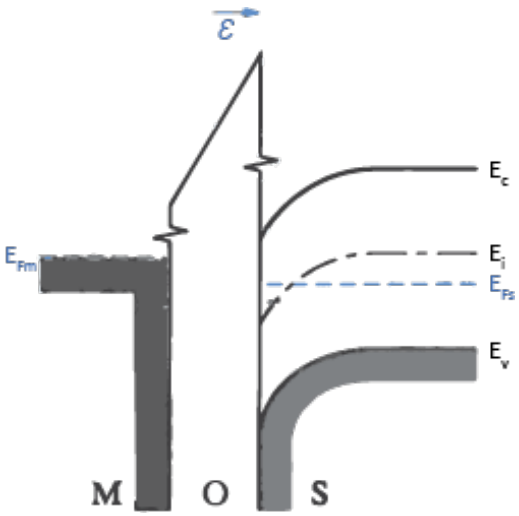
$$C_i = \frac{\epsilon_i}{d} = \frac{3.9\times8.85\times10^{-14}}{1\times10^{-6}} = 3.45\times10^{-7}\ F/cm^2$$

$$Q_d = -qN_aW_m = -1.6\times10^{-19}\times1\times10^{16}\times3.0\times10^{-5}$$

$$Q_d = -4.8\times10^{-8}\ C/cm^2$$

$$\boldsymbol{V_T} = \Phi_{ms} - \frac{Q_i}{C_i} - \frac{Q_d}{C_i} + 2\phi_F$$

$$\boldsymbol{V_T} = -1.0 - \frac{2.5\times10^{-8}}{3.45\times10^{-7}} + \frac{4.8\times10^{-8}}{3.45\times10^{-7}} + 2\times0.347 = \mathbf{-0.24}\ \boldsymbol{V}$$



c) Calculate the drain current at $V_G$ = 5V and $V_D$ = 0.1V.

For $V_G$ = 5V, $V_D$ = 0.1V and $V_T$ = -0.24V: $V_D < (V_G - V_T)$ → we are in the linear regime.

$$I_D = \frac{Z}{L}\overline{\mu_n}\, C_i[(V_G - V_T)V_D - \frac{1}{2}V_D^2\ ]$$

$$I_D = \frac{25}{1}200\ \cdot 3.45 \cdot 10^{-7}\ [(5 + 0.24)0.1 - \frac{1}{2}0.1^2\ ]$$

$$\boldsymbol{I_D = 0.90mA}$$

d) Calculate the drain current at $V_G$ = 3V and $V_D$ = 5V.

For $V_G$ = 3V, $V_D$ = 5V and $V_T$ = -0.24V: $V_D > (V_G - V_T)$ → we are in the saturation regime.

$$V_D(sat.) = V_G - V_T = 3\ + 0.24 = 3.24\ V$$

$$I_D = \frac{Z}{L}\overline{\mu_n}\, C_i[(V_G - V_T)V_D(sat.) - \frac{1}{2}V_D(sat.)^2\ ]$$

$$I_D = \frac{25}{1}200\ \cdot 3.45 \cdot 10^{-7}\ [(3 + 0.24)3.24 - \frac{1}{2}3.24^2\ ]$$

$$\boldsymbol{I_D = 9.1\ mA}$$

e) Discuss what would happen if $V_G$ were to be kept at 3V and $V_D$ was increased to 7V.

Increasing $V_D$ further will not change the current notably (that is, until breakdown occur at much higher voltages) as we are in saturation. The additional voltage drop would build up over the depletion region.

# Form for "Multiple Choice"

(Tear off this form, mark your answers and hand in that page together with the rest of your exam.)

Candidate number:____LF_____

| Alternative: Question number.: | a | b | c |
|---|---|---|---|
| II-1 | | | **x** |
| II-2 | **x** | | |
| II-3 | **x** | | |
| II-4 | | **x** | |
| II-5 | | **x** | |
| II-6 | | | **x** |
| II-7 | **x** | | |
| II-8 | **x** | | |
| II-9 | | **x** | |
| II-10 | | **x** | |